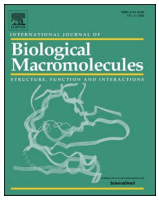
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The semi-synthesis, characterization, and antiviral properties of alginate-based nitrated derivatives

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1. Introduction

Alginate is a linear polysaccharide consisting of 1 → 4 linked β-D-mannuronic acid (M) and α-L-guluronic acid (G) in the form of M-blocks (polymannuronic acid, PM), G-blocks (polyguluronic acid, PG), and interspersed MG-blocks [1]. Owing to its biocompatibility, biodegradability, nontoxicity, stability, hydrophilicity, and gelling property, alginate has been extensively employed for a variety of biomedical purposes [2]. To expand its application, alginate is always used as a backbone for semi-synthesizing novel macromolecules with functional substituents. These functional substituents in turn enable to tailor alginate's properties [3,4]. Although alginate is the only natural polysaccharide with one carboxyl group per monosaccharide residue, which brings highly strong polyanionic property, natural alginate has no obvious antiviral property. In order to enhance its antiviral property, sulfated modification of alginate is the commonly used strategy. After sulfation, the resulting alginate sulfate shows broad-spectrum antiviral activities against both DNA and RNA viruses, such as HPV [5], HIV [6], HBV [7], and SARS-CoV-2 [8]. Recently, due to the high incidence of viral infections, the development of safe and effective antiviral drugs or biomaterials is still vital, and novel alginate-based antiviral agents are worth bing investigated.

Nitric oxide (NO) is a signaling molecule produced in mammalian cells by NO synthase, and is involved in various physiological processes [9]. It is widely recognized that NO plays a key role in the pathogenesis of many human viral infections [10], and possesses direct or indirect antiviral activities [11,12]. NO can exert antiviral effects through direct inhalation [13]. What is more noteworthy is that NO can be indirectly administered through NO donors, achieving controlled and sustained NO release. Many NO donors with different structures and NO release principles have been designed to realize antiviral effects in different scenarios [14,15].

Polysaccharide nitrates, such as starch nitrate, cellulose nitrate, and chitosan nitrate, are one kind of inorganic polysaccharide esters [16]. It's currently known that these nitrated derivatives are generally water-insoluble and susceptible to explosion, which limits their applications in the pharmaceutical and tissue engineering industries. Like polyhydroxy nitrates, such as glyceryl trintrate (GTN), pentaerythrityl tetranitrate (PETN) [17], polysaccharide nitrates can act as NO donors, in which the nitrate groups endow the NO release characteristics. The NO release abilities of polysaccharide nitrates should be exploited for therapeutic applications. In theory, alginate can react with several types of inorganic acids to produce inorganic alginate esters. Alginate sulfate and phosphate are the common semi-synthetic derivatives [18–20]. Alginate nitrate, as one kind of inorganic alginate esters, there is no systematic report regarding the preparation method, physicochemical property, and biological function. It is worth investigating whether alginate nitrate, as a combination of alginate and NO, can serve as a potential antiviral agent. The PM and PG have different structures in alginate [21,22], which results in different physicochemical properties and bioactivities. In our previous studies, we have found that the backbones of PM and PG affected the bioactivities of alginate-based derivatives largely [19,20]. It is important to study the properties of PM and PG side by side, when novel alginate-based derivatives are produced. This strategy can help to design new derivatives with different M/G ratio to target different purposes. Therefore, the nitrated derivatives of low-molecular-weight PM (LPM) and low-molecular weight PG (LPG) were prepared as NO donors side by side in this study. What changes obtained after nitration? Did alginate-based nitrated derivatives as NO donors exhibit potential antiviral activities? What differences of

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physicochemical properties and antiviral activities occurred between LPM nitrate (LPMN) and LPG nitrate (LPGN)? Systematic studies were conducted in this study.

2. Experimental

2.1. Reagents

Alginate was purchased from Qingdao Gather Great Ocean Algae Industry Group Co., Ltd. (Qingdao, China). All chemical reagents used in this study were of analytical grade. Vero and RAW264.7 cells were purchased from the Cell Bank of the Institute of Biochemistry and Cell Biology, Chinese Academy of Sciences (Shanghai, China). Mouse monoclonal antibodies against vesicular stomatitis virus G (VSV-G) were purchased from Santa Cruz Biotechnology (CA, USA), and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was purchased from Cell Signaling Technology (Massachusetts, USA). Fetal bovine serum (FBS) was purchased from Gbico Corporation (Life Technologies Corporation, USA). DMEM high glucose medium was purchased from Zhejiang Senrui Biotechnology Co., Ltd. (Zhejiang, China).

2.2. Preparation of LPMN and LPGN

Alginate was degraded using 0.5 mol/L HCl at 100 °C for 10 h, and then the obtained precipitate was dissolved in 2 % Na_2CO_3 solution. The solution was adjusted to pH 2.85 using HCl, and then centrifugated to obtain supernatant (crude PM) and precipitate (crude PG). The supernatant (crude PM) was hydrolyzed at 110 °C for 20 min, and then adjusted to pH 2.80. The mixture was centrifugated, and the resulting supernatant was desalinated and spray-dried to obtain LPM. The precipitate (crude PG) was resuspended in distilled water, and adjusted to pH 2.85. The mixture was hydrolyzed at 120 °C for 1 h, and then adjusted to pH 2.90. The mixture was centrifugated, and the resulting precipitate was redissolved in 2 % Na_2CO_3 followed by desalinating and spray drying to obtain LPG. LPMN and LPGN were prepared according to the HNO_3/P_2O_5 method [16]. 1 g of P_2O_5 was added slowly to 10 mL of fuming nitric acid (95 %) with stirring in an ice bath for 1 h, and then the undissolving P_2O_5 was filtered out to obtain HNO_3/H_3PO_4 mixture. Subsequently, 1 g of LPM or LPG powder was added, and the mixture was stirred for 1 h in an ice bath. The reaction mixture was then precipitated with CH_2Cl_2. The obtained precipitate was collected and washed with acetone three times before being redissolved in distilled water. The obtained water solution was neutralized to pH 7–8 using NaOH, dialyzed, rotary evaporated and freeze-dried to obtain LPMN or LPGN.

2.3. Analytical methods

The nitrogen content (N%) and carbon content (C%) were determined according to the method of elemental analysis using a Vario EL III element analyzer (Elementar, Germany). The DS_N, which represented the average number of nitrate group (-NO_2) on each mannuronic acid or guluronic acid residue, was calculated based on the ratio of N% and C% (DS_N = 36 N%/7C%). The weight-average molecular weight (Mw) and number-average molecular weigh (Mn) were determined using high efficiency gel permeation chromatograph (HPGPC) on an Agilent 1100 (Agilent, USA) using our previously reported method [23], and the polydispersity (PD) was calculated as Mw/Mn. The UV–visible (UV–Vis) spectra were determined on a METASH UV-6000PC spectrophotometer (Shanghai Metash Instruments Co.,Ltd., China) at 190–900 nm wavelengths. The fourier transform infrared (FT-IR) spectra were conducted on a Nicolet Nexus 470 FT-IR spectrometer (Thermo, USA) using the KBr pellet technique. All nuclear magnetic resonance (NMR) spectra were collected on a JNM-ECP600M spectrometer (JEOL, Japan). The Congo red binding experiment was conducted based on a previously reported method [24]. The specific rotation [α] were collected on a Rudolph/Autopol-I polarimeter (Rudolph, USA) at concentration of 1 % (1 dm measurement tube, sodium D line 589.3 nm, 20 °C). The thermo gravimetric analysis (TGA) data were analyzed on the model q600 (10 °C/min, nitrogen protection) using a TGA Q600 thermal gravimetric analyzer (Netzsch, Germany).

2.4. NO release assay

2.4.1. NO release in the presence of L-cysteine

The NO release assay was conducted based on the previously reported method [25] with some modifications. Briefly, LPMN and LPGN were dissolved in phosphate buffer solution (PBS, pH 7.4) or PBS containing L-cysteine (3.4 mmol/L) to obtain 0.1 mg/mL mixtures. The mixtures were incubated at 37 °C. 500 μL of each mixture was taken at different time points, and then 500 μL of Greiss reagent (the mixture of 1 % sulfanilic acid, 0.1 % N-1-naphthylethylenediamine dihydrochloride, and 6 % phosphoric acid) was added. The obtained mixtures were kept at 37 °C for 10 min, and the absorbance of each mixture was read at 540 nm. Solutions of 0–200 μmol/L sodium nitrite were employed to prepare a nitrite absorbance *versus* concentration curve under the same experimental conditions. The NO release ability was calculated based on the standard nitrite concentration curve.

2.4.2. NO release in the cell culture medium

LPMN and LPGN were dissolved in DMEM medium (containing 10 % FBS and without phenol red) to obtain 0.1 mg/mL mixtures. The mixtures were incubated at 37 °C for 5 h. The subsequent steps were executed according to method describing in part 2.4.1.

2.5. Anti-VSV assay in Vero cell in vitro

2.5.1. Cell viability assay

Vero cells were plated into 96-well sterile polystyrene plates (3.5×10^3 cells/well). The Vero cells were incubated with LPMN or LPGN for 48 h. Thereafter, the resazurin was added to each well at a resazurin/medium ratio of 1:10 (v/v). After incubation at 37 °C for 4 h, the absorbance was read at 544/595 nm using a SpectraMax M3 reader (Molecular Devices, Sonny Wilber, USA).

2.5.2. VSV inhibition assay

The VSV expressing a green fluorescence protein (GFP) was propagated and purified following the standard protocol [26]. Briefly, the VSV with the serum-free medium was inoculated at MOI = 0.01 for 30 min. Later, the culture media of cells was replaced with fresh media. After infection for 12 h, a fluorescence microscope was used (Carl Zeiss Axio Vert.A1, Gottingen, Germany) to observe GFP expression, and then the cells were harvested. The Vero cells were pretreated with LPMN or LPGN at indicated concentrations, or IFNβ (10 ng/mL) for 4 h. Then, the Vero cells were incubated with VSV (MOI = 0.01) for 30 min. After virus infection, the Vero cells were incubated with LPMN, LPGN or IFNβ for 12 h, respectively. Bright-field and GFP fluorescence of Vero cells were imaged (Scale bar: 100 μm). The expression of the VSV-G protein was measured by ImageJ software. The data represented results from at least three independent experiments. NTC was the no treatment control.

2.5.3. Western blotting assay

The Vero cells were harvested and lysed using the cell lysis buffer, including 50 mmol/L Tris pH 7.6, 150 mmol/L NaCl, 0.5 % Triton X-100, 1 mmol/L Na_3VO_4, 10 mmol/L NaF, 5 mmol/L Na-pyrophosphate, 10 mmol/L β-glycerophate, PMSF and protease inhibitor cocktails. Equal amounts of total protein were subjected to SDS-PAGE and transferred onto nitrocellulose membranes. After blockade with 5 % milk (w/v), the membranes were incubated with antibodies against VSV-G and GAPDH at 4 °C overnight. Then, the membranes were washed by TBST with 0.1 % tween-20 (v/v). Finally, the membranes were immunoblotted with HRP-conjugated secondary antibodies (1:5000 dilutions) at

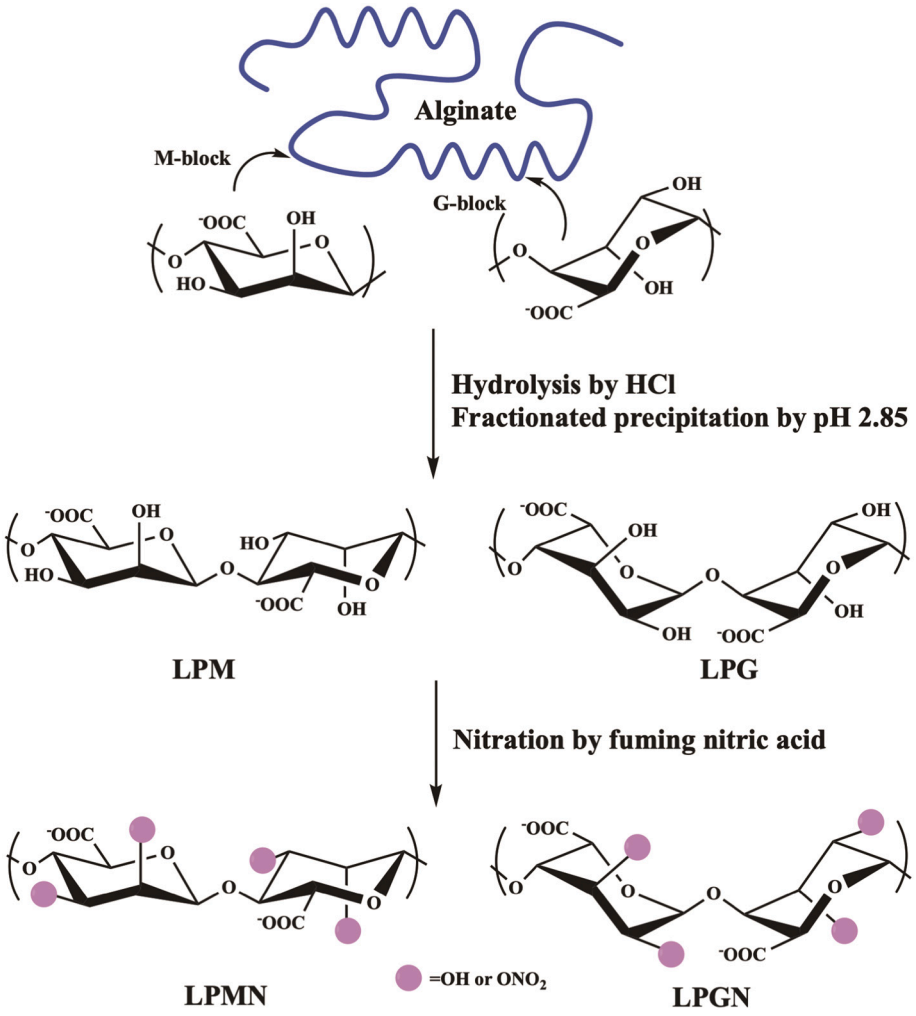


Fig. 1. The semi-synthesis route of LPMN and LPGN.

room temperature for 1 h. All membranes were visualized by using the enhanced chemiluminescence (ECL) western blotting reagent.

2.6. Transcriptome assay in Raw264.7 cell in vitro

The RAW264.7 cells were treated with LPGN or LPMN (800 ng/mL) for 6 h. Then, total RNA was extracted for transcriptome analysis. Transcriptome data were analyzed by OE Biotechnology Co., Ltd., China. *P*-values were adjusted using the Benjamini-Hochberg method to control the false discovery rate, with $p \leq 0.05$ regarded as the threshold for significant differential expression. The protein-protein interaction (PPI) network was analyzed using Metascape (https://metascape.org/) [27].

2.7. Anti-VSV assay in vivo

C57BL/6 mice were purchased from Charles River Laboratories (Beijing, China). C57BL/6 mice were infected with VSV (5×10^7 PFU per gram of body weight) *via* intravenous injection. Then, the mice were treated with PBS or LPGN *via* intraperitoneal injection for 7 day. All animal experiments were conducted in accordance with the "National Institutes of Health Guidelines for the Care and Use of Laboratory Animals" and approved by the Animal Ethics Committee of the University of Health and Rehabilitation Sciences (KFDX:NO.2025-1001).

Total RNA was extracted from tissues using the TRIzol reagent (Invitrogen, CA, USA) according to the manufacturer's protocol. cDNA was synthesized from 1 μg RNA using a cDNA synthesis kit (Vazyme, Nanjing, China) according to the instructions. Real-time PCR was conducted using a ChamQ Universal SYBR qPCR Master Mix (Vazyme, Nanjing, China) in a Light Cycler 480 (Roche, Basel, Switzerland). The PCR protocol was: initial activation at 95 °C for 5 min, 40 cycles at 95 °C for 15 s, and 60 °C for 1 min. Ct values were converted into relative gene expression levels, and compared to that of the internal control genes 18s using the 2-ΔCt method. VSV N/P-F, CAAAACAGAAAACCGACTCC, VSV N/P-R, AGACGAGAATAGGACTTGAGA; 18s-F, ATTGACGGAAGGGCACCACCAG, 18s-R, CAAATCGCTCCACCAACTAAGAACG.

2.8. Statistical analyses

The data were presented as mean ± SD. Statistical analysis was performed using ANOVA and *t*-test.The GraphPad Prism 8.0 (San Diego, CA, USA) was utilized for graph generation.

3. Results and discussions

3.1. Semi-synthesis of LPMN and LPGN

The preparation route of LPMN and LPGN is shown in Fig. 1. In alginate, the glycosidic bond of the interspersed MG-block can be more easily hydrolyzed compared to that of M-block and G-block [1]. PM and PG can be released based on the partial acid hydrolysis of alginate in strong acid heterogeneous conditions. In this study, LPM and LPG were

Table 1
Characteristics of LPMN and LPGN.

Compound	N%	C%	DS_N	Mw (kDa)	PD	Specific rotation [α]
LPM	–	–	–	6.8	1.18	−86.02
LPG	–	–	–	7.0	1.30	−105.32
LPMN	–	–	–	5.8	1.26	−51.81
LPGN	8.03	21.85	1.89	6.4	1.38	−92.39

prepared according to the further degradation of the pH-fractionated hydrolyzed-alginate. The fuming nitric acid (HNO_3 > 90 %) is the mostly used nitration reagent in the semi-synthesis of polysaccharide nitrate esters, and always paired with the catalysts such as P_2O_5, $(CH_3CO)_2O$, and H_2SO_4 [16]. The HNO_3/P_2O_5 method is generally used in the nitration of easily acid-hydrolyzed polysaccharides. In this method, P_2O_5 turns into H_3PO_4 after absorbing the residual water of the fuming nitric acid, and the H_3PO_4 in turn acts as a catalyzer for nitration. To minimize the degradation of LPM and LPG, the HNO_3/P_2O_5 method was used to prepare the water-soluble nitrated derivatives of LPM and LPG in an ice bath, by which their C2-OH and C3-OH were substituted by $-NO_2$. Compared to the nitration of water-insoluble polysaccharides, it is challenging to isolate the nitrated products of soluble polysaccharides from a huge amount of acid. We found that dichloromethane (CH_2Cl_2) could be used to precipitate the nitrated products of LPM and LPG from the reaction mixtures. The obtained precipitates were collected, washed, dissolved, neutralized, dialyzed and freeze-dried to obtain LPMN and LPGN, respectively.

3.2. Characterization of LPMN and LPGN

Alginate is the only polysaccharide naturally containing a carboxyl group in each constituent residue. Therefore, the maximum DS_N of alginate-based nitrated derivative is 2. The N% of LPMN was 7.22 % with a DS_N of 1.80, and the N% was 8.03 with a DS_N of 1.89 for LPGN, as shown in Table 1. These data suggested that most of hydroxyls were substituted by $-NO_2$.

In HPGPC analysis, the Mw was 6.8 kDa for LPM, and 7.0 kDa for LPG. It was inevitable that the strong acidic reaction conditions led to the degradation of the nitrated products. The Mw of LPMN and LPGN were 5.8 and 6.4 kDa, respectively. The Mw decreased 1.0 kDa from LPM to LPMN, and 0.6 kDa from LPG to LPGN, which suggested that the backbone of PG was more resistant to acid hydrolysis than that of PM. After nitration, the molecular weight distribution of LPMN and LPGN became greater than that of LPM and LPG, as shown in Fig. 2A.

UV–Vis spectra were conducted at concentration of 1 mg/mL under wavelengths ranging from 190 to 900 nm, as shown in Fig. 2B and C. The characteristic absorption wavelengths of LPM and LPG were all around 190 nm, which was the characteristic absorption of polysaccharides. After nitration, the maximum absorption peaks at 220 nm and the shoulder peaks at 280 nm confirmed the presence of $-NO_2$.

Fig. 2D shows the FT-IR spectra. Compared to LMP and LPG, three

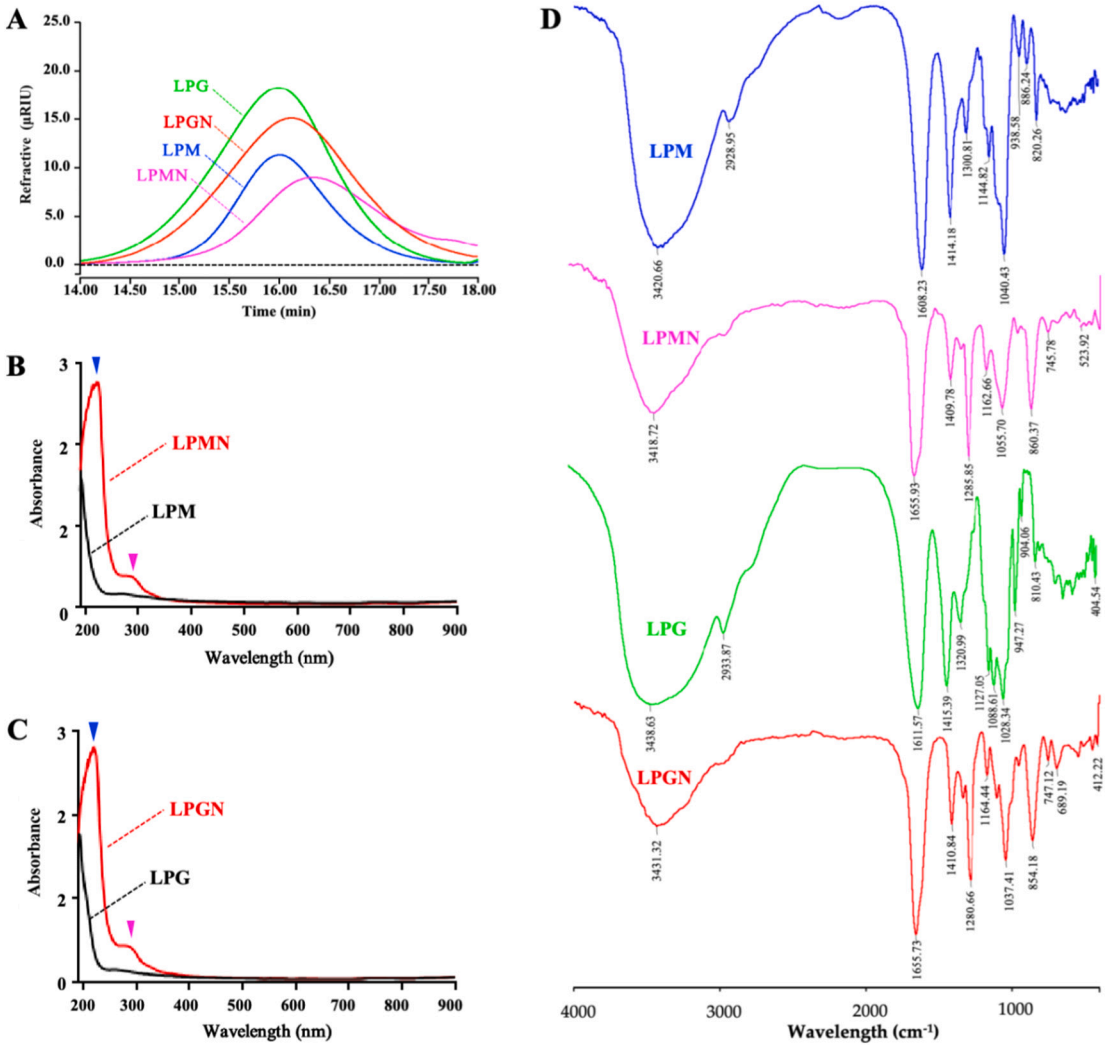


Fig. 2. The HPGPC chromatogram (A), UV–Vis spectra (B and C) and FT-IR spectra (D).

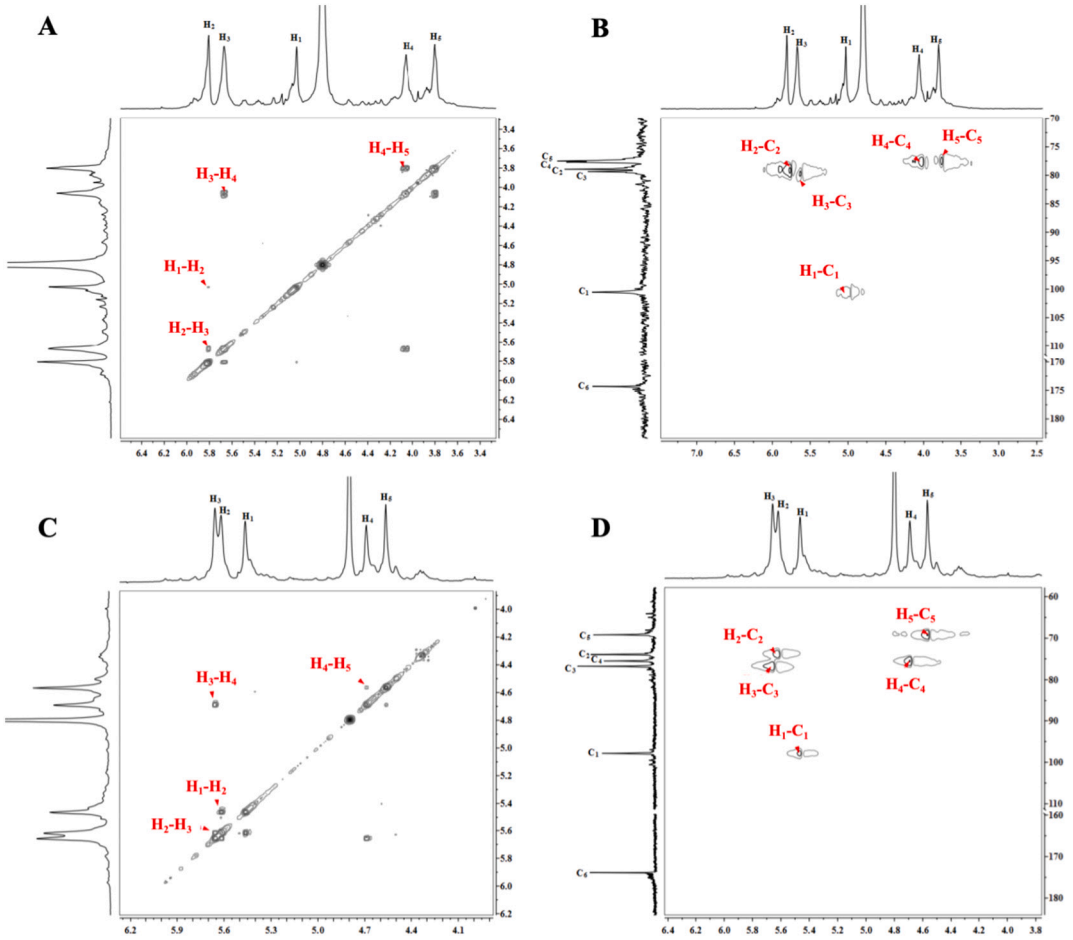


Fig. 3. The NMR spectra of LPMN (A as ^{1}H, ^{1}H COSY spectrum, and B as ^{1}H, ^{13}C HSQC spectrum) and LPGN (C as ^{1}H, ^{1}H COSY spectrum, and D as ^{1}H, ^{13}C HSQC spectrum).

Table 2
The ^{1}H NMR and ^{13}C NMR chemical shifts of LPMN and LPGN.

Compound	^{1}H NMR (ppm)[a]					
	H1	H2	H3	H4	H5	
LPMN	5.03	5.81	5.67	4.06	3.80	
LPGN	5.45	5.62	5.66	4.69	4.57	
	^{13}C NMR (ppm)[b]					
	C1	C2	C3	C4	C5	C6
LPMN	100.84	79.24	79.65	78.01	77.79	174.65
LPGN	98.18	74.24	77.10	75.80	69.50	174.20

[a] The ^{1}H NMR spectra were recorded with H_2O signal as standard (δ-OH = 4.80 ppm, 25 °C).
[b] The ^{13}C NMR spectra were recorded with the acetone as an internal standard (δ-CH_3 = 31.45 ppm).

new intense bands of LPMN/LPGN at 1655.93/1655.73 cm^{-1}, 1285.85/1280.66 cm^{-1}, and 860.37/854.18 cm^{-1} attributed to the antisymmetric -NO_2 stretching, the symmetric -NO_2 stretching, and the valence -NO stretching, respectively [28]. The bands at 1409.78/1410.84 cm^{-1} were corresponded to the symmetric carboxyl (-COOH) stretching. The bands for antisymmetric -COOH stretching, which should lie around 1690 cm^{-1} [23], were covered in the antisymmetric stretching of -NO_2.

The ^{1}H NMR and ^{13}C NMR spectra of LPMN and LPGN were analyzed based on the ^{1}H, ^{1}H COSY and ^{1}H, ^{13}C HSQC spectra (Fig. 3), and the NMR signal assignments are listed in Table 2. Since most of hydroxyls were substituted by -NO_2, both the ^{1}H NMR and ^{13}C NMR spectra of LPMN and LPGN exhibited distinct peaks. It had been reported that the chemical shifts of C1, C2, C3, C4, C5, and C6 were observed to be 101.80, 71.86, 73.21, 79.63, 77.73, and 177.11 ppm for PM, and 102.88, 66.84, 70.94, 82.14, 69.07, and 177.50 ppm for PG [19]. -NO_2 is a strong electron acceptor. After the substitution of C2-OH and C3-OH by -NO_2, the signals of C2 and C3 in the ^{13}C NMR spectra of LPMN and LPGN markedly shifted downfield compared to those of LPM and LPG. The signals of unsubstituted C2 and C3 almost disappeared. Affected by -NO_2, the signals of H2 and H3 shifted to the lower field regions than H1 in the ^{1}H NMR spectra of LPMN and LPGN.

In order to further understand the spatial-structural characteristics of LPMN and LPGN, Congo red binding experiments were conducted, as shown in Fig. 4A. Congo red is an acidic dye that can form complexes with polysaccharides with triple helix conformations. The maximum absorption wavelength of the complex will red shifted compared to Congo red [24]. When NaOH concentration exceeds a certain value, the maximum absorption wavelength decreases. Compared to LPM and LPG, LPMN and LPGN showed weak redshift phenomenons, indicating that the original helical structures no longer existed after nitration. Furthermore, nitrated structural modification reduced the left-handed rotation state compared to the raw materials of LPM and LPG, as shown in Table 1.

Nitrated derivatives of polyhydroxy compounds will become unstable along with the increase of DS_N. Trinitroglycerin (DS_N = 3), starch nitrate (DS_N = 3), and cellulose nitrate (DS_N = 2.45–2.87) can be used as explosives [16]. As nitrate esters, the thermal stabilities of LPMN and LPGN were analyzed using TGA. As shown in Fig. 4B, LPMN and LPGN all showed process of instant-weight-loss compared to LPM and LPG.

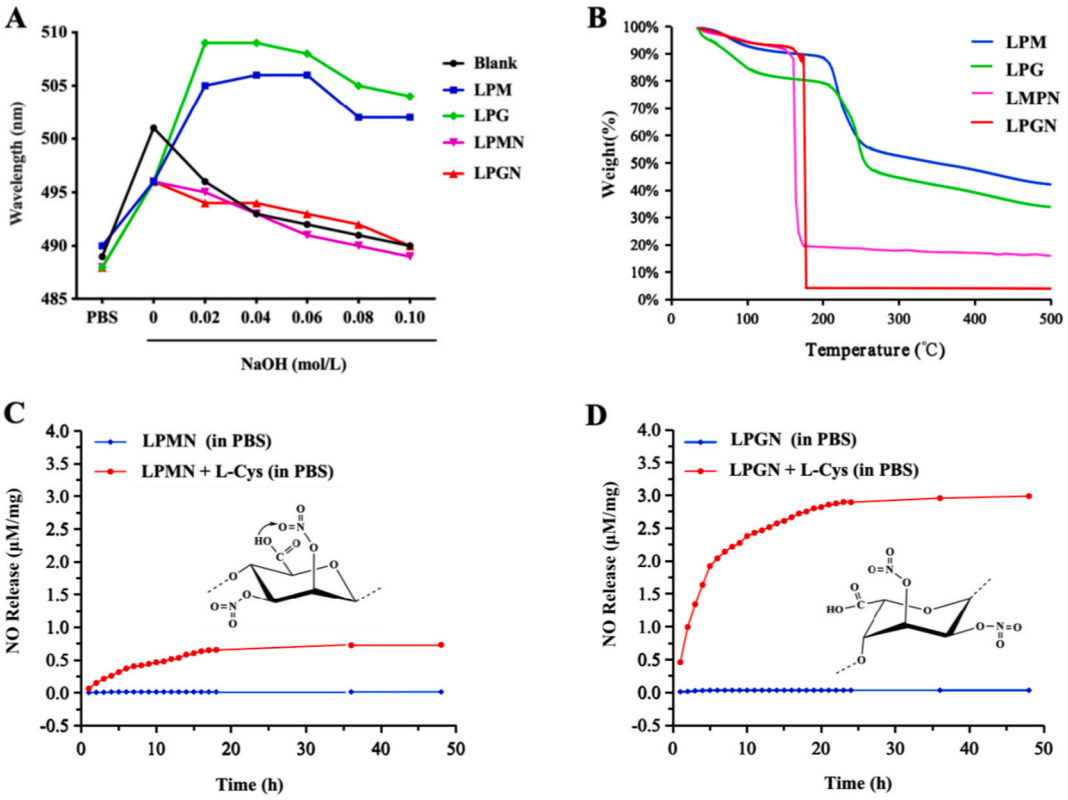


Fig. 4. The Congo red binding experiments (A), TGA data (B), and NO release trend of LPMN (C) and LPGN (D).

The temperature range of the process was 164–174 °C for LPMN, and 175–178 °C for LPGN. It was speculated that the process of instant-weight-loss was associated with the deflagration, like other nitrate esters. The trigger deflagration temperature of LPGN was relatively higher than that of LPMN, suggesting that LPGN was more stable than LPMN.

3.3. NO release of LPMN and LPGN

NO is an essential cellular signaling molecule involving in many physiological and pathological processes [9]. Therefore, strategies for delivering exogenous NO hold promise for a number of biomedical applications. Some small-molecular organic nitrates, such as glyceryl trinitrate (GTN), isosorbide mononitrate (ISMN), and pentaerythrityl tetranitrate (PETN), have been employed as NO donor drugs in clinical settings. LPMN and LPGN, as macromolecules, also belong to organic nitrates. In order to find out if LPMN and LPGN could act as NO donors, and if the backbones of LPMN and LPGN affected NO release, we measured their NO release abilities *in vitro*.

NO release assay in the presence of L-cysteine is a classic method for evaluating polyhydroxy nitrates. In this assay, NO releases from the nitrosothiols, which is an intermediate of the nitrate group and L-cysteine, and then rapidly transforms to NO_2^- [29]. After various incubation times in the absence/presence of L-cysteine at 37 °C, aliquots were added to Griess reagent to quantify the released NO in the form of NO_2^- [25]. As shown in Fig. 4C and D, both LPMN and LPGN exhibited no NO release after incubation in PBS, suggesting that they were stable in the absence of L-cysteine. Upon incubation with L-cysteine, significant NO release was observed in LPMN (Fig. 4C) and LPGN (Fig. 4D). Based on the NO release rate, the process could be divided into three phases: phase I from 1 h to 5 h, phase II from 6 h to 17 h (LPMN) or 23 h (LPGN), and phase III to the end of 48 h. The quantity and percentage of NO release at the end point of each phase are summarized in Table 3. Both LPMN and LPGN exhibited rapid NO release in the phase I. The NO release percentages were 4.96 % for LPMN, and 30.77 % for LPGN after 5 h. Along with the incubating time, the NO release rate slowed down in phase II. In phase III, little NO released, even though LPMN and LPGN were incubated for >48 h.

Table 3
NO release data of LPMN and LPGN.

Compound	5 h		17 h /23 h		48 h	
	a	p (%)	a	p (%)	a	p (%)
LPMN*	0.32	4.96	0.65	10.07	0.73	11.32
LPMN#	0.28	4.34	–	–	–	–
LPGN*	1.92	30.77	2.90	46.47	2.98	47.76
LPGN#	1.66	26.60	–	–	–	–

a: the actual NO release at the end point of each phase, μmol/mg. p: the NO release percentage at the end point of each phase, which was calculated by (actual NO release)/(theoretic NO release). *: the tested sample in PBS in the presence of L-cysteine. #: the tested sample in DMEM medium containing 10 % FBS.

Polyhydroxy nitrates can release NO in blood due to the presence of some proteins containing sulfhydryl groups [30]. As shown in Table 3, both LPMN and LPGN released NO in DMEM medium containing 10 % FBS. The NO release percentages were 4.34 % for LPMN and 26.60 % for LPGN after 5 h, which was slightly lower than that in the presence of L-cysteine.

LPGN showed significantly better NO release capacity than LPMN. Since LPMN and LPGN had similar DS_N, there may be special structures to prevent the NO release in LPMN. The Mw analysis revealed that LPMN and LPGN had relatively short chains of about 20 M or G units. The Congo red experiments showed that both LPMN and LPGN have no obvious triple helical structure in PBS. The -NO_2 of LPMN and LPGN may not all participate in forming regular secondary structures, thus reducing the NO release capacity. As shown in Fig. 4C, the -COOH and -NO_2 of LPMN were located on the same face of the mannuronic acid ring. We speculated that a hydrogen bond formed between the two

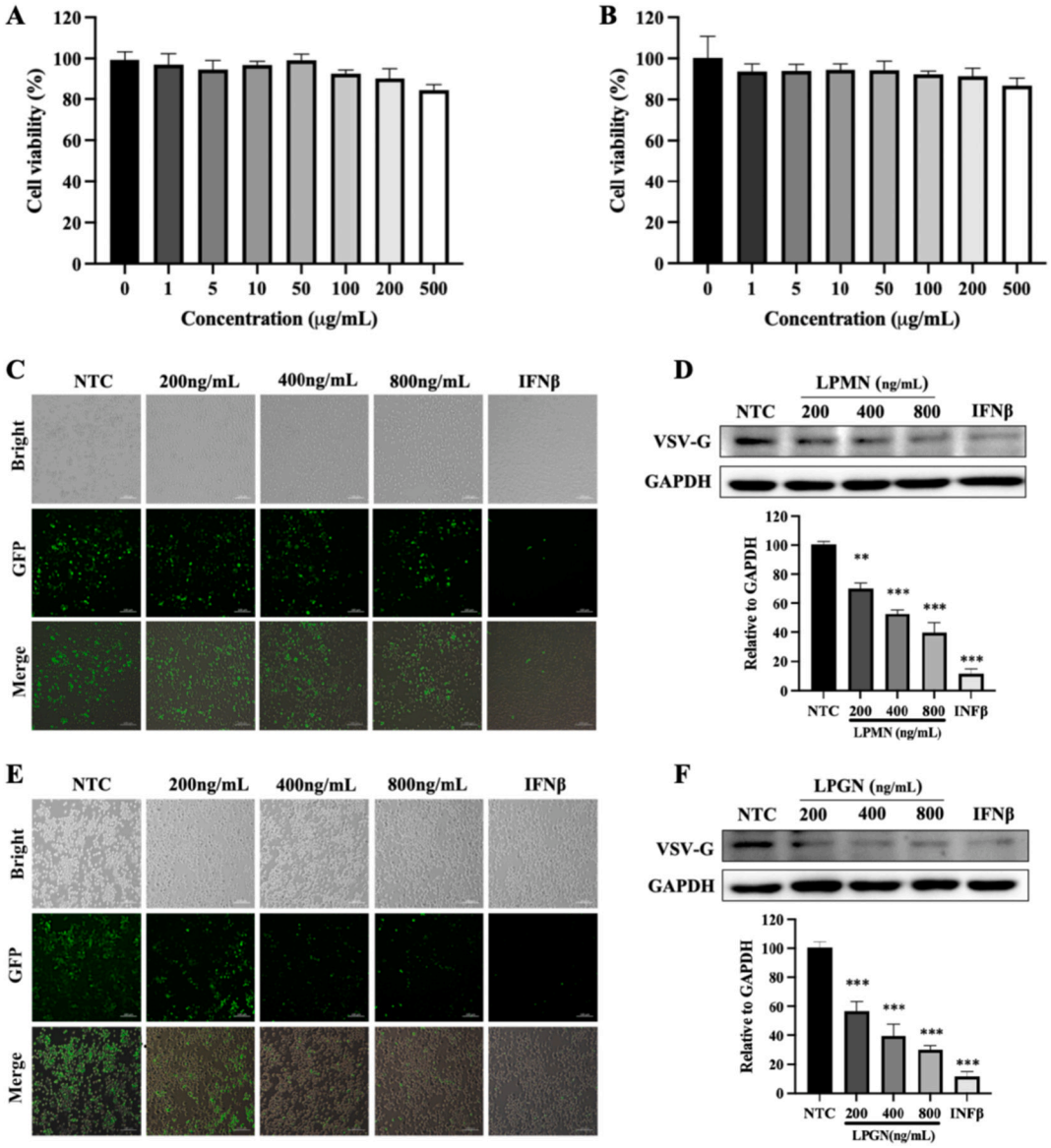


Fig. 5. The cell viability of Vero cells treated with LPMN (A) and LPGN (B), and the anti-VSV properties of LPMN (C as the bright field and GFP fluorescence, and D as the expression of VSV-G protein) and LPGN (E as the bright field and GFP fluorescence, and F as the expression of VSV-G protein) *in vitro*. The data represent results from three independent experiments. ** $p < 0.01$ compared to the NTC group; *** $p < 0.001$ compared to the NTC group.

groups, which prevented -NO_2 binding with L-cysteine or sulfhydryl group to form the nitrosothiols. In LPGN, hydrogen bond may hardly be formed between -COOH and -NO_2 in the guluronic acid ring, as shown Fig. 4D.

Compared to small-molecular NO donors, the development of macromolecular NO donors is a burgeoning field with great therapeutic usages owing to the tunable NO release characteristics. In this study, LPMN and LPGN showed different NO release capacities, suggesting that alginate-based nitrated derivatives can be used as potential NO release biopolymers, in which the M/G ratio can control the NO release. The stability of NO donor is an important property. In Fig. 4B, the physical stabilities of both LPMN and LPGN were affected by high temperatures. In this part, LPMN and LPGN all release NO in the presence of L-cysteine, which revealed on the other hand that the substance containing sulfhydryl group was the factor affecting their chemical stabilities.

3.4. Anti-VSV properties of LPMN and LPGN in vitro

Among inorganic polysaccharide esters, sulfated derivatives are most commonly studied for their excellent heparin-like activities. In our previous work, we have demonstrated that the sulfate group was the key substituent for the heparin-like activities compared to phosphate and H-phosphonate groups in alginate-based derivatives [19]. As a similar situation, the sulfate group is also the key substituent for showing the antiviral activities of sulfated polysaccharide, such alginate sulfate [5–8], glucan sulfate [31], fucoidan [32], and carrageenan [32]. The sulfate group is probably necessary to bind with the positive charges on the surface of virus [33]. For nitrate group, as a neutral group, there is no report about the charge-binding ability to virus. Despite being a polyanionic polysaccharide, alginate has no obvious antiviral activity. Therefore, in this study, the antiviral activities of LPMN and LPGN *in vitro* should be mainly related to the released NO.

The Vero cell model is considered an ideal cell model for studying direct antiviral activity [34]. Before evaluating antiviral activities, the toxicities of LPMN and LPGN on Vero cells were tested. The cell viability of Vero cells after co-culturing with different concentrations of LPMN or LPGN solution for 48 h are shown in Fig. 5A and B. The results showed that neither LPMN nor LPGN led to significant cytotoxicity (cell

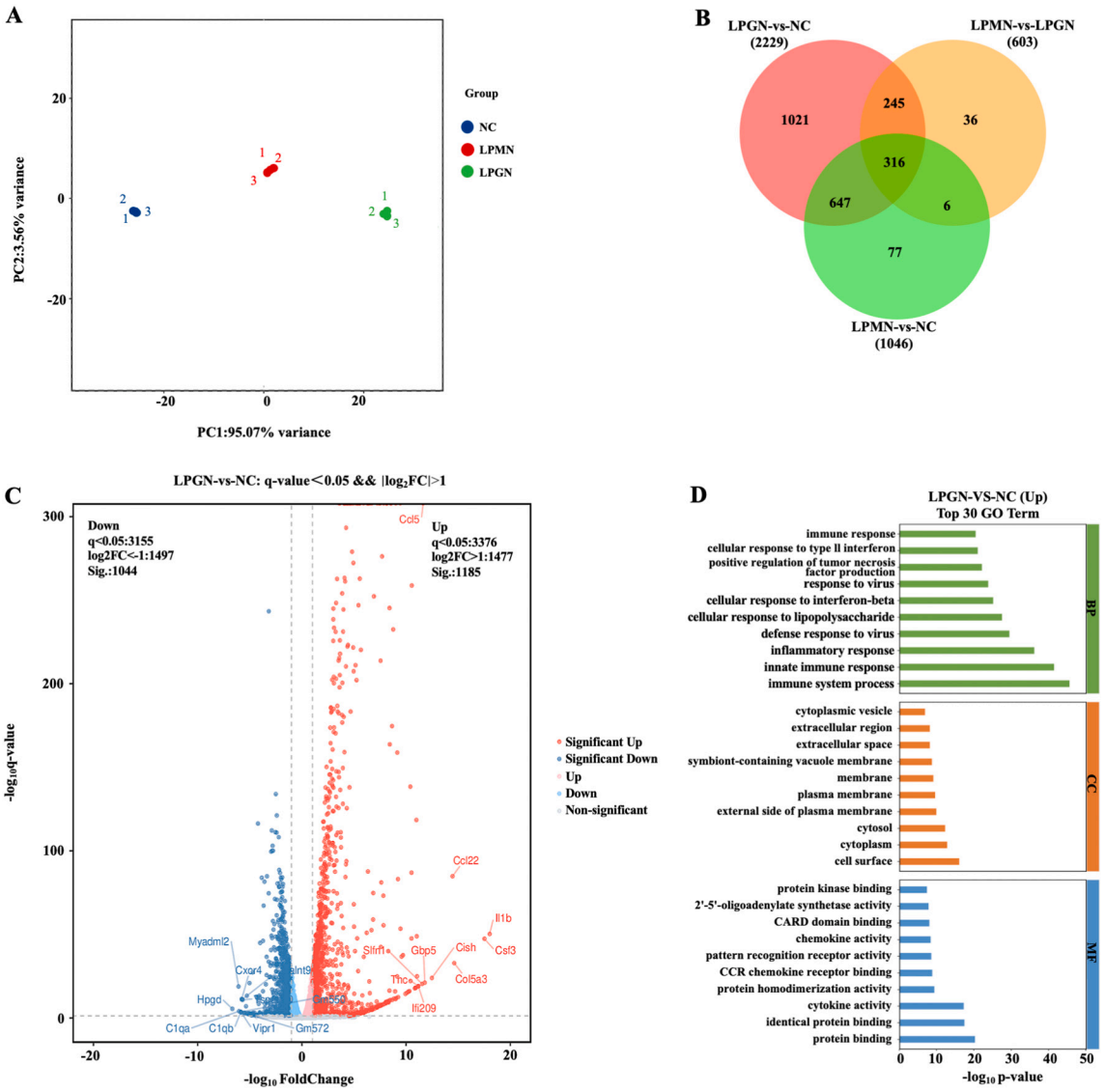


Fig. 6. Transcriptome analysis for LPMN and LPGN. (A) Principal Component Analysis. (B) A venn diagram illustrating the expression of different genes. (C) Volcano maps of significantly different genes between the NC and LPGN groups. (D) GO function annotation histogram of significantly different genes in the NC and LPGN groups.

viability>80 %) in the measured concentration range (0–500 μg/mL). To assess the antiviral activities of LPMN and LPGN, we employed an engineered green fluorescent VSV reporter strain, and the GFP reporter gene fused to the previously reported VSV-GFP [26]. According to using the VSV-GFP, observing green fluorescence could directly evaluate the VSV replication. As shown in Fig. 5C and E, the fluorescence of GFP gradually reduced as the concentration of LPMN or LPGN increasing in Vero cells. These data suggested that LPMN or LPGN inhibited VSV viral replication. The VSV genome encodes five major proteins [34], including matrix protein (M), nucleoprotein (N), large polymerase protein (L), phosphoprotein (P), and glycoprotein (G). The VSV-G protein represents the amount of VSV in Vero cells. The VSV-G protein was significantly suppressed after treatment with LPMN and LPGN at a low does, as shown in Fig. 5D and F. These data revealed that LPMN and LPGN resisted the infection of the VSV. It was worthy to note that LPGN showed better anti-VSV property than LPMN, which was mainly related to the higher NO release capacity of LPGN than that of LPMN. These data also suggested that alginate-based nitrated derivatives with high G content would have greater potential to be designed as antiviral biopolymers.

3.5. Immunoregulation properties of LPMN and LPGN in vitro

In the Vero cell model *in vitro*, LPMN and LPGN directly inhibited the VSV proliferation. As we know, antiviral infection can also be achieved by regulating the immune system *in vivo*. In our previous work, LPMN and LPGN have displayed the immune activation activities in splenocyte and macrophage model *in vitro* [20]. To further explore the immunoregulation properties of LPMN and LPGN, which could demonstrate the indirect antiviral abilities, transcriptome was employed to investigate differential gene expressions in RAW264.7.

As shown in Fig. 6A, the principal component analysis (PCA) confirmed the quality and suitability of NC, LPMN, and LPGN groups for subsequent analysis. Through these data, it could be observed that NC, LPMN, and LPGN groups formed clear clusters. PCA data suggested that LPMN and LPGN treatment significantly altered the gene expression profile compared to the NC group. The venn diagram revealed 2229 differentially expressed genes (DEGs) in the LPGN group compared to the NC group, with LPMN showing 1046 DEGs compared to the NC group, as shown in Fig. 6B. However, the LPGN group only had 603 DEGs compared to the LPMN group. These data suggested that the cells

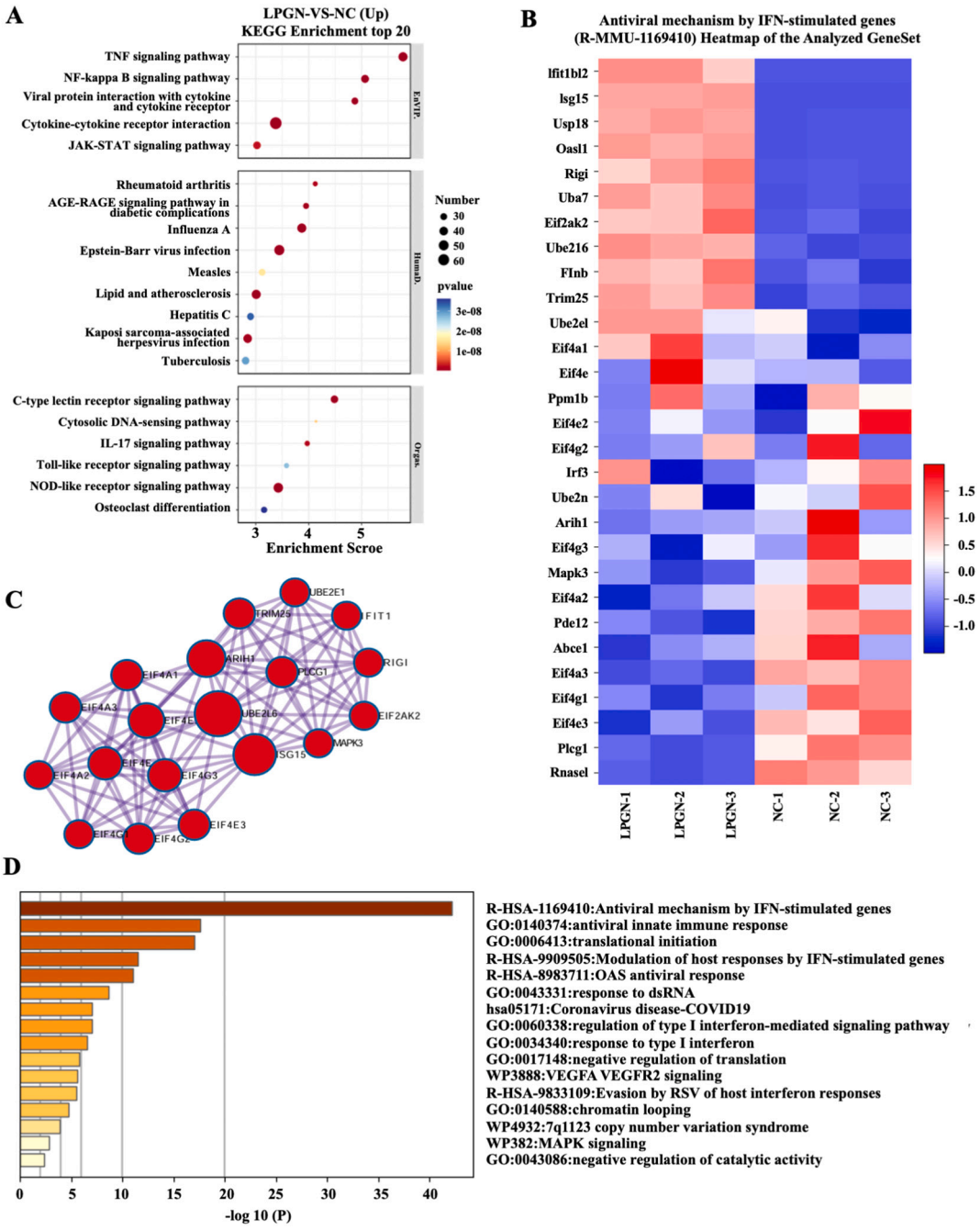


Fig. 7. Transcriptome analysis for LPMN and LPGN. (E) A bubble map illustrating the KEGG channel enrichment analysis of potential targets for LPGN group. (F) Heat maps of significantly different genes in the heart tissues of the NC and LPGN groups. (G) Protein-protein interaction network identified in the gene lists. (H) Bar graph of enriched terms across input gene lists, colored by *p*-values.

of the LPGN and LPMN groups shared similar signaling pathways.

As LPGN shown more DEGs (*vs.* NC) and batter anti-VSV activity *in vitro* than those of LPMN, the further studies were conducted for LPGN. In Fig. 6C, volcano plot visualization identified 1185 upregulated and 1044 downregulated genes in LPGN-treated cells compared to the NC group. The genes upregulated by LPGN were predominantly associated with immune modulation, including key mediators, such as il1b, ccl22, csf3, ccl5. Gene ontology (GO) analysis shown in Fig. 6D also corroborated these findings, with enriched terms clustering in immune response regulation (biological process) and cytokine activity (molecular function).

Interferon (IFN), mainly being secreted by immune cells, is an important cytokine with antiviral properties [35]. In Fig. 7A, KEGG pathway analysis highlighted some significantly enriched pathways, such as viral protein-cytokine interactions, C-type lectin receptor signaling, and NOD-like receptor signaling pathways. Of particular interest was the marked upregulation of interferon-stimulated genes (ISGs) including ifit1bl2, ISG15, and UPS18, as evidenced by hierarchical clustering analysis, as shown in Fig. 7B and C. This observation, coupled with pathway enrichment results in Fig. 7D, suggested that the antiviral activity of LPGN could be mediated through type I IFN activation. We hypothesized that LPGN could potentially engage pattern recognition receptors to initiate innate immune responses, thereby conferring antiviral protection.

However, this study now has no direct evidence to explain if the immunoregulation properties of alginate-based nitrated derivatives

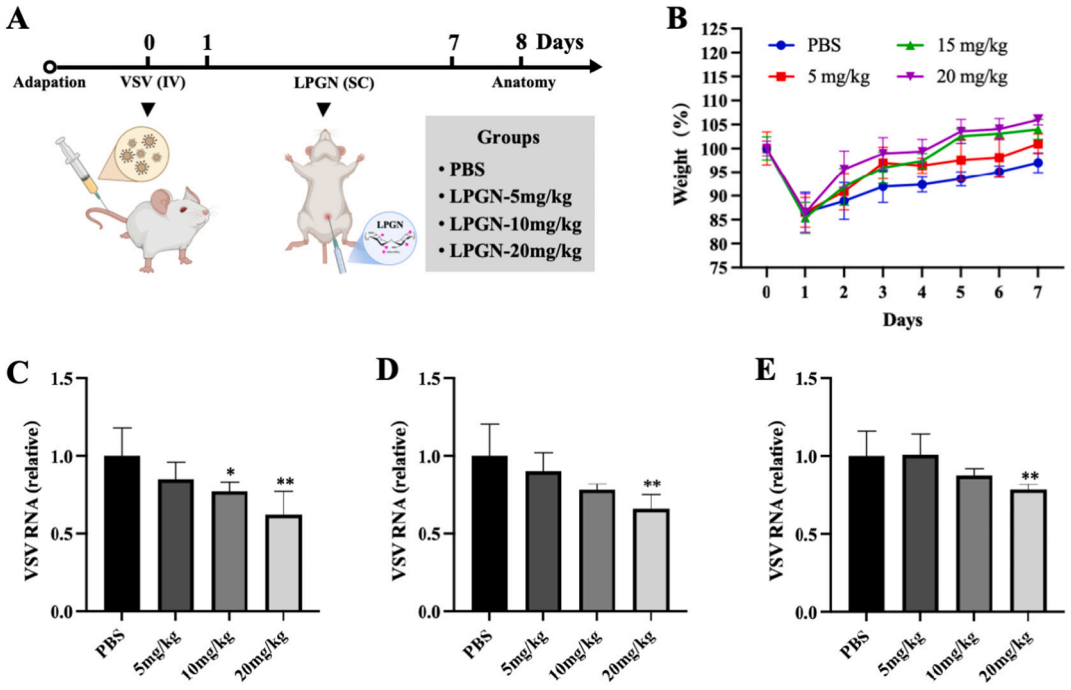


Fig. 8. LPGN inhibited VSV *in vivo*. (A) C57/B6 mice were infected with VSV (5×10^7 PFU per gram of body weight) *via* intravenous injection, then the models was treated with PBS or LPGN *via* intraperitoneal injection for 7 day. (B) The weight of VSV infected mice. VSV-N/P mRNA in kidney (C), liver (D), and lung (E). mRNA levels were quantified by quantitative real-time PCR and normalized to 18s mRNA levels. Values are means ± S.D. ($n = 5$). * $p < 0.05$ compared to the PBS group; ** $p < 0.01$ compared to the PBS group.

related to the released NO, or the nitrated derivatives itself, or both the two factors. To further explore the alginate-based nitrated derivatives, more mechanism research will be evaluated in the future.

3.6. Anti-VSV properties of LPGN in vivo

As LPGN showed anti-VSV and immunoregulation properties *in vitro*, we evaluated the anti-VSV property *in vivo* using a mouse model of VSV infection, as shown in Fig. 8A. After infection with VSV, the weight of the mice rapidly decreased and gradually recovered (Fig. 8B). The body weight of LPGN-treated mice increased gradually and existed dose-dependent manner, suggesting that the LPGN was safety *in vivo*. The VSV-N/P mRNA levels in kidney, liver, and lung were analyzed using qRT-PCR. The data in Fig. 8C-E showed that LPGN significantly and dose-dependently inhibited VSV *in vivo*.

4. Conclusions

In this study, the new nitrated derivatives of LPM and LPG with high DS_N were prepared as novel alginate-based NO donors using the HNO_3/P_2O_5 method, and their characteristics and structures were systematically elucidated. After nitration, new characteristics appeared in LPMN and LPGN. LPMN and LPGN were stable in PBS, while released NO in the presence of L-cystein and DMEM medium containing FBS. Due to the structural reasons, LPGN had better NO release property than LPMN. Both LPMN and LPGN showed anti-VSV and immunoregulation properties *in vitro*, and LPGN had better properties. LPGN was further verified the anti-VSV activity *in vivo*. All these findings in this study provide an new idea for designing alginate-based nitrated derivatives, especially for tailoring their M/G ratio to obtain controlled NO donors and antiviral biopolymers.

CRediT authorship contribution statement

Quancai Li: Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Wenwei Han:** Writing – original draft, Visualization, Software, Methodology, Investigation, Funding acquisition, Formal analysis. **Chunxia Li:** Resources, Methodology, Investigation, Funding acquisition. **Jiangying Sun:** Validation, Software, Data curation. **Depeng Lv:** Validation, Data curation. **Meng Shao:** Validation, Data curation. **Fusheng Sun:** Formal analysis.

Declaration of competing interest

The authors report no relationships that could be construed as a conflict of interest.

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Data availability

Data will be made available on request.

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